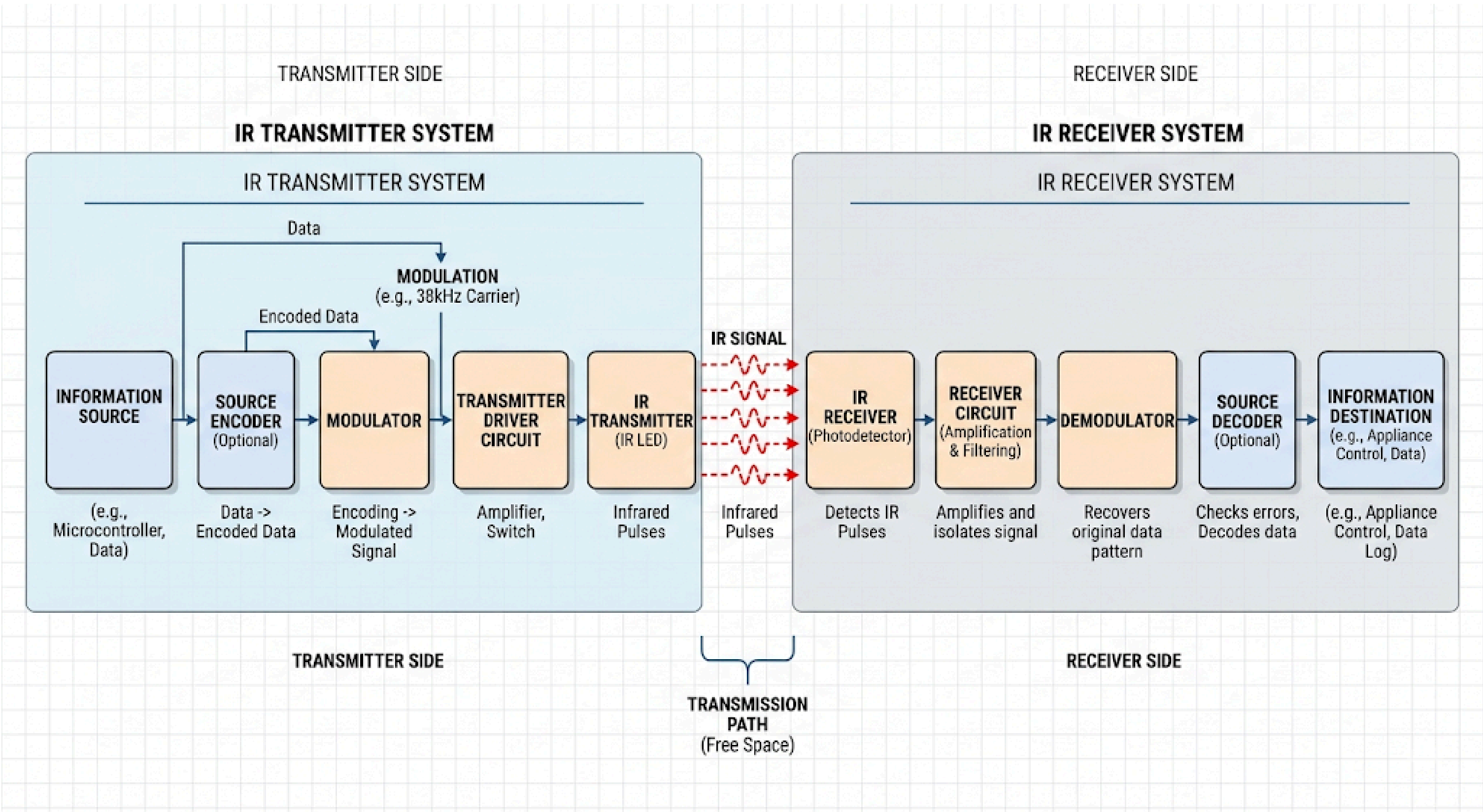
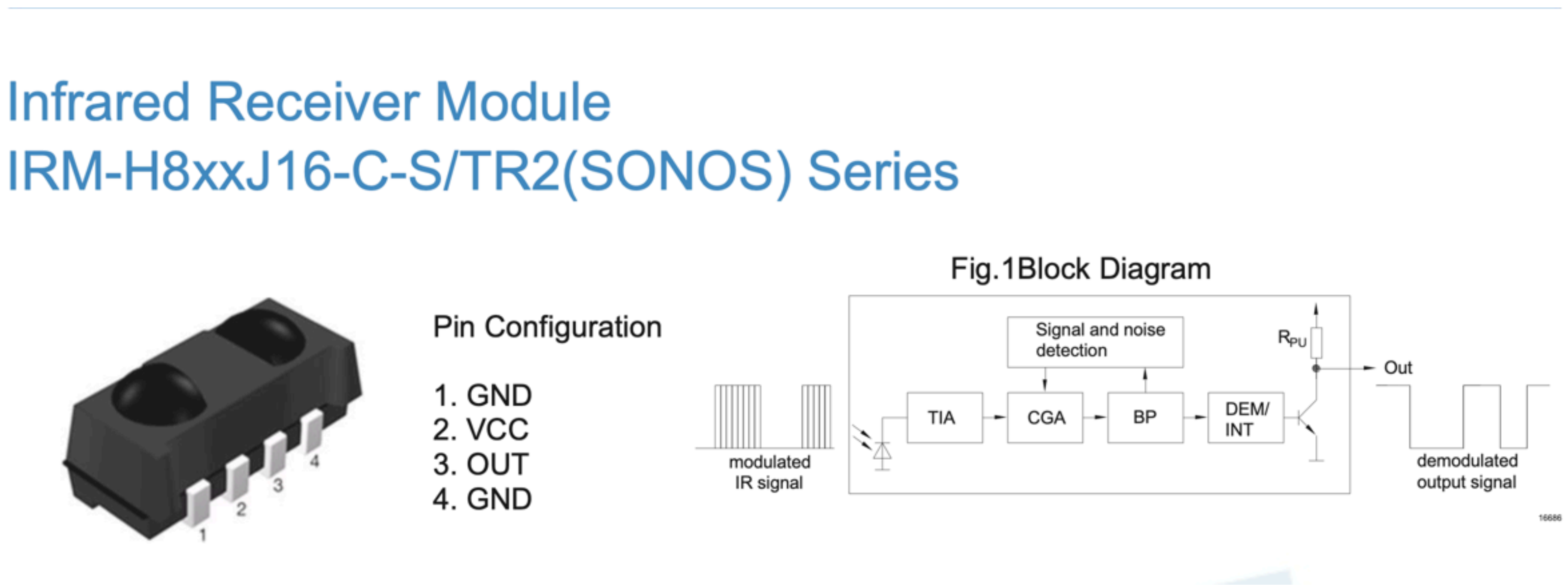
IR System Test Study

IR block principle and diagram



Sonos used everlight IR receiver



1. **The Input Stage**

- **Modulated IR Signal:** The process begins on the left side of the diagram when a "modulated IR signal" (the blinking infrared light from a remote) hits the receiver.
- **PIN Diode & Filter:** Although not explicitly labeled with text in the diagram, the diode symbol represents a PIN diode that captures the light. The physical module itself is molded in black epoxy that acts as an initial IR filter to block out unwanted light.

2. **The Processing Core** Once the light is converted into an electrical signal, it travels through several internal processing blocks:

- **TIA & CGA (Amplification and Gain Control):** The signal first enters the preamplifier stages. While the acronyms aren't strictly defined in the text, the sources describe an **Adaptive Gain Control (AGC)** function that adjusts the gain to suppress optical noise from the environment. This correlates to the **CGA** (Controlled Gain Amplifier) block.
- **BP (Band-pass Filter):** The signal then moves to the **BP** block. This is the internal **Band-pass Filter**, which is responsible for distinguishing the remote control's specific carrier frequency (such as 38kHz) from other ambient light flicker, like that from CFL bulbs.
- **Signal and Noise Detection:** This block runs in parallel to the main signal path, helping the module's logic ignore chaotic IR "bursts" (like those from a plasma TV) so that it only processes actual remote commands.
- **DEM/INT (Demodulator):** Finally, the signal reaches this block to be demodulated. It extracts the raw data from the carrier frequency so it can be outputted as a simple on/off digital pulse.

3. **The Output Stage**

- *RPU* **(Pull-up Resistor):** Before leaving the module, the signal passes an internal pull-up resistor (*RPU*), which the datasheet notes has a typical value of 45 kΩ.
- **Demodulated Output Signal:** The final result at the "Out" pin is a clean, "demodulated output signal" (shown as a square wave in the diagram) that can be directly decoded by a microprocessor.

UPDATED IR SYSEM VALIDATION PLAN

Sonos uses IR receiver datasheet:

tsop15300.pdf IRM-H8xxJ16-C-S-TR2(SONOS) Series Datasheet V2 (1).pdf